Movie Genre Classification from Plot Summaries Using TF-IDF and Logistic Regression

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***Abstract*—Movie genre classification is an important task in natural language processing and recommendation systems. In this project, we investigate the use of textual plot summaries for predicting movie genres. Each movie description is first transformed into a numerical representation using the TF-IDF feature extraction method. A logistic regression model is then trained to perform multi-class genre classification. A key challenge in this task is the presence of class imbalance, where some genres appear much more frequently than others. To address this issue, we explore two strategies: logistic regression with class weighting and logistic regression trained on oversampled data. The performance of the models is evaluated using accuracy, macro F1-score, and confusion matrix analysis. Experimental results show that incorporating class weights significantly improves macro F1-score compared to the baseline model, indicating better performance on minority genres. Feature importance analysis further reveals that genre-related keywords strongly influence model predictions. Overall, the results demonstrate that TF-IDF features combined with logistic regression provide an effective and interpretable baseline for movie genre classification.**

***Index Terms*—genre classification, natural language processing, TF-IDF, logistic regression, class imbalance**

I. Introduction

Automatic movie genre classification has become an increasingly important task in natural language processing and multimedia information retrieval. With the rapid growth of online streaming platforms and digital media libraries, large numbers of movies are produced and distributed every year, making manual genre tagging inefficient and time-consuming. Automatically predicting movie genres from textual descriptions such as plot summaries can help efficiency in large movie databases.

Prior work has explored a variety of machine learning approaches for predicting movie genres. For example, Hoang (2018) investigated several machine learning models, including Naïve Bayes, XGBoost, and recurrent neural networks, to classify movie genres based on plot summaries. The study demonstrated that textual information contained in movie synopses can provide useful signals for genre prediction, and that plot neural network models can achieve competitive performance on large movie datasets [1].

More recent work has explored multimodal approaches that combine multiple sources of information. Mangolin et al. (2022) proposed a multimodal framework that integrates textual data, movie posters, subtitles, and trailer videos to perform multi-label movie genre classification. Their results show that combining different data sources can improve classification performance because each modality captures complementary information about movies [2]. However, textual features such as plot summaries remain one of the most accessible and informative sources for genre prediction.

Recent research has also explored deep learning models that focus specifically on textual plot descriptions. For example, Shaukat et al. (2025) proposed a transformer-based architecture with a hierarchical attention mechanism to perform multi-label movie genre classification using movie plots. Their model leverages contextual language representations to capture semantic relationships within narrative text and demonstrates strong performance across several benchmark datasets [3]. These findings highlight the growing potential of text-based deep learning models for genre classification tasks.

Another important challenge in movie genre classification is multi-label and class imbalance. Movies often belong to multiple genres simultaneously, and some genres appear much more frequently than others. This imbalance may bias machine learning models toward majority classes and reduce prediction performance for less frequent genres.

In this project, we investigate movie genre classification using textual plot summaries and traditional natural language processing techniques. Specifically, we employ TF-IDF features to represent textual data and apply logistic regression models to perform genre prediction. In addition to building a baseline classifier, we explore strategies to address class imbalance, including class-weighted logistic regression and random oversampling, to improve classification performance across different genres. Through experiments on a movie plot dataset, we evaluate how different modeling strategies influence classification performance and analyze which textual features contribute most strongly to genre prediction.

II. Method

A. Problem Formulation

This project investigates whether a movie's genre can be predicted from its plot summary. The task is formulated as a multi-class text classification problem.

Each movie in the dataset includes a short plot description in the overview field. This text serves as the input to the model, while the movie genre serves as the prediction target. The goal is to learn a model that can identify the genre of movie based on its textual description.

Formally, let x_i denote the plot summary of movie i, and let y_i denote its genre label. The learning task is to estimate a function:

$$f(x_i) \rightarrow y_i \tag{1}$$

that maps the movie description to the corresponding genre category.

Because machine learning models require numerical inputs, the text must first be converted into feature vectors. In this project, TF-IDF vectorization is used to represent the textual information contained in the movie summaries.

B. Data Description

The dataset used in this project is the TMDB 5000 Movie Dataset, obtained from Kaggle. It contains metadata and textual information for thousands of movies.

Two variables are central to this analysis. The overview column provides a short description of the movie plot, which serves as the input text for the classification task. The genres column contains the genre information associated with each movie.

In the original dataset, a movie may belong to multiple genres. To simplify the classification problem, the first genre listed for each movie is extracted and used as the primary genre label.

An initial inspection of the dataset shows that the distribution of genre is highly imbalanced, with some genres appearing much more frequently than others. This imbalance is an important factor to consider when training and evaluating the classification models.

C. Text Representation

Since machine learning models require numerical inputs, the textual plot summaries must first be converted into numerical features. In this project, the Term Frequency—Inverse Document Frequency (TF-IDF) method is used to represent the movie descriptions as vectors.

TF-IDF is a widely used technique in text classification that assigns weights to words based on their importance in a document and their frequency across the entire dataset. Words that appear frequently in a document but rarely other documents receive higher weights, which helps highlight terms for classification [4].

Using the vector space representation, each document can be written as a vector:

$$d_i = (w_1, w_2, w_3, \ldots, w_n) \tag{2}$$

where w_j represents the weight of term j on document i.

The TF-IDF weight of a term is computed as:

$$TFIDF(t, d) = TF(t, d) \times IDF(t) \tag{3}$$

where $TF(t, d)$ is the frequency of term t in document d, and $IDF(t)$ measures how rare the term is across the entire corpus.

After applying TF-IDF vectorization, each movie plot summary is transformed into a numerical feature vector, which is then used as the input for the classification.

D. Classification Model

After converting the plot summaries into TF-IDF feature vectors, a Logistic Regression model is used for genre classification.

Logistic regression is a commonly used method for classification tasks. It estimates the probability that a sample belongs to a particular class based on the input features. The model uses the logistic function:

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}} \tag{4}$$

where x represents the TF-IDF feature vector and denotes the model parameters.

For each movie description, the model predicts the probability of different genres. The genre with the highest predicted is selected as the final prediction.

To address class imbalance in the dataset, two additional strategies are explored: logistic regression with class weights and logistic regression trained on oversampled data.

E. Model Evaluation

To evaluate the performance of the classification model, accuracy, precision, recall and F1-score are used. Accuracy measures the overall proportion of correct predictions. Because the genre distribution is imbalanced, F1-score is especially important, as it balances precision and recall and provides a more informative measure of model performance across genres.

III. Results

A. Baseline Model Performance

We first trained a baseline logistic regression classifier using TF-IDF features extracted from movie plot summaries. The model achieves an accuracy of 0.433 and a macro F1-score of 0.217 on the test set.

Despite the relatively low macro F1-score, the model shows reasonable performance for several genres. For example, Comedy achieves an F1-score of 0.55, while Action and Drama reach F1-scores of 0.47 and 0.49, respectively. These results suggest that the model can capture meaningful textual patterns associated with certain movie genres.

However, the classification report also reveals substantial performance differences across genres. Categories such as Animation, Crime, and Thriller receive very low F1-scores, indicating that the model struggles to correctly identify these less frequent classes.

The confusion matrix in Fig.1 further illustrates this pattern. The model frequently predicts genres such as Drama and Comedy, while minority genres are rarely predicted correctly. Many minority genres are misclassified as Drama or Comedy, which are the dominant classes in the dataset. This observation highlights the impact of class imbalance in the dataset and motivates the strategies explored in the following section.

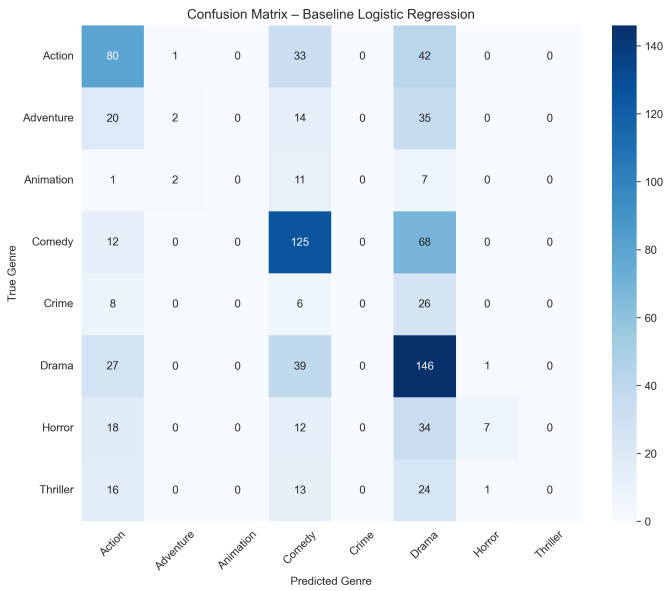

Fig. 1. Confusion Matrix of the Baseline Logistic Regression Model.

B. Addressing Class Imbalance

To address the imbalance observed in the baseline model, two strategies were tested. The first method applies class-weighted logistic regression, where the parameter class_weight = "balance" increases the importance of minority classes during training. The second method uses random oversampling, which balances the training data by duplicating samples from minority genres before fitting the model.

Table I compares the performance of the baseline model and the two imbalance-handling methods. While the accuracy increases only slightly (from 0.433 to about 0.46), the macro F1-score improves substantially, rising from 0.217 to 0.373 with class weighting and 0.363 with oversampling. Since macro F1gives equal importance to all classes, this improvement indicates that the model becomes better at recognizing minority genres.

Overall, both methods help alleviate the impact of class imbalance, with class-weight logistic regression achieving the best overall performance.

C. Feature Importance Analysis

To better understand how the classifier distinguishes between movie genres, we analyze the coefficients of the trained logistic regression model. Since TF-IDF features are used to represent the plot summaries, the model coefficients indicate how strongly each word contributes to predicting a particular genre.

For each genre, the words with the largest positive coefficients are extracted as the most influential features. These keywords provide insight into the textual patterns learned by the model.

Several meaningful patterns can be observed. For example, Action and Crime movies are both associated with words related to law enforcement and violence, such as cop, agent, police, murder, and killer. The overlap of these terms suggests that the model captures a common vocabulary describing criminal activities, which may also explain why these genres can sometimes be confused.

Other genres show more distinctive vocabulary. Comedy movies often contain relationship-related words such as love, wedding, and friends, while Horror films include characteristic terms like zombies, terrifying, and demonic. These words provide strong signals for the classifier and help explain why some genres are easier to identify than others.

TABLE I
COMPARISON OF MODEL PERFORMANCE

Model	Accuracy	Macro F1
Baseline Logistic Regression	0.433	0.217
Logistic Regression (Class Weight)	0.463	0.373
Logistic Regression (Oversampling)	0.461	0.363

IV. CONCLUSIONS

In this project, we investigated the task of movie genre classification using textual plot summaries and traditional machine learning methods. TF-IDF features were used to represent textual information, and logistic regression models were applied to predict movie genres.

The baseline logistic regression model achieved an accuracy of 0.433 and a macro F1-score of 0.217. To address class imbalance, class-weighted logistic regression and random oversampling were further tested. The best performance was obtained by class-weighted logistic regression, which improved accuracy to 0.463 and macro F1-score to 0.373, while the oversampling model achieved an accuracy of 0.461 and a macro F1-score of 0.363. These results show that handling class imbalance substantially improves performance across genres, especially across minority classes. In addition, the feature importance analysis showed that the classifier learned meaningful genre-related keywords from plot summaries.

However, this study has several limitations. First, although movies often belong to multiple genres, this study simplifies the task by predicting only a single genre label for each movie. This simplification may ignore important genre combinations such as "Action-Adventure" or "Comedy-Drama". Second, the model relies only on textual plot summaries, while other useful information such as cast information, or release year was not considered. Finally, the TF-IDF representation combined with logistic regression is a relatively simple linear model that may not fully capture deeper semantic relationships in text.

Future work could explore more advanced deep learning approaches for movie genre classification. For example, neural models such as recurrent neural networks or transformer-based architectures could be applied to learn contextual representations directly from plot summaries rather than relying on TF-IDF features. These models may better capture semantic relationships and long-range dependencies within movie descriptions, which could improve the ability to distinguish

between similar genres and enhance predictions for minority classes. In addition, future research could implement a true multi-label classification framework so that each movie can be assigned to multiple genres simultaneously. Incorporating additional movie metadata, such as cast information, keywords, or release year, may also provide complementary signals that further improve genre prediction performance.

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